

Data Cleaning, Transformation & Visualization with Power BI



Project Objective:

This project focuses on **data transformation, cleaning, modeling, and visualization in Power BI** to generate meaningful business insights from sales data. The workflow includes importing multiple datasets, performing ETL operations, creating relationships between tables, developing calculated fields using DAX, and designing an interactive dashboard for decision-making.



Dataset Description:

This dataset contains sales, customer, product, and order-related information used to perform data transformation, data modeling, and business analysis in Power BI. The purpose of the dataset is to analyze business performance, identify sales trends, evaluate profitability, and generate actionable insights through interactive dashboards.

The dataset includes multiple related tables such as:

- **Orders Table** – Contains order details including Order ID, Order Date, Customer ID, Product ID, Quantity, Sales, and Discount.
- **Customers Table** – Stores customer information such as Customer ID, Customer Name, Segment, Region, and other demographic details.
- **Products Table** – Includes product-related details like Product ID, Product Name, Category, and Sub-Category.
- **Sales/Profit Table** – Contains revenue, cost, and profit-related metrics used for financial analysis.
- **Targets Table (if applicable)** – Stores sales targets for performance comparison and KPI tracking.

Key Objectives of the Dataset

- Clean and transform raw business data using ETL processes.
- Build relationships between multiple tables through data modeling.
- Create calculated columns and DAX measures.
- Analyze sales performance and profitability.
- Develop interactive dashboards and reports for business decision-making.

Tools Used

- **Power BI Desktop** – Data Modeling & Dashboard Development
- **Power Query Editor** – Data Cleaning & Transformation
- **DAX (Data Analysis Expressions)** – Calculations and KPI creation



Column Description

Column Name	Description
Order ID	Unique identifier for each order
Order Date	Date when the order was placed
Customer ID	Unique identifier for each customer
Customer Name	Name of the customer
Product ID	Unique identifier for each product
Product Name	Name of the purchased product
Category	Product category (Electronics, Clothing, etc.)
Quantity	Number of units purchased
Unit Price	Price per unit of the product
Discount	Discount applied to the order
Revenue	Total sales amount generated
Cost	Total cost of products sold
Profit	Revenue minus cost
Region	Geographic sales region
Sales Target	Expected sales value



Data Cleaning & Transformation

- Imported raw datasets into Power BI and verified data types.
- Removed duplicate records to maintain data accuracy.
- Handled missing values using replacement rules and transformations.
- Standardized text fields (customer names, categories, IDs).
- Corrected inconsistent formats and renamed columns for clarity.
- Created calculated columns and measures using DAX where required.
- Established relationships between tables for efficient data modeling.
- Applied transformations using Power Query Editor:
 - **Change Data Types**

1	B-2	168	
2	B-2		
3	B-2		
4	B-2		
5	B-25602	168	

Split/Merge Columns

Merge Columns		
Choose how to merge the selected columns.		
Separator --None--		
New column name (optional) Merged		
OK Cancel		

State	City	Location
Gujarat	Ahmedabad	Ahmedabad,Gujarat
Maharashtra	Pune	Pune,Maharashtra
Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh
Rajasthan	Jaipur	Jaipur,Rajasthan
West Bengal	Kolkata	Kolkata,West Bengal
Karnataka	Bangalore	Bangalore,Karnataka
Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir
Tamil Nadu	Chennai	Chennai,Tamil Nadu
Uttar Pradesh	Lucknow	Lucknow,Uttar Pradesh

Conditional Columns

Add a conditional column named Profit Status based on:

- Profit < 0 → Loss
- Profit = 0 → Break-Even
- Profit > 0 → Profit

Transform Add Column View Tools Help

Conditional Column Index Column Duplicate Column Merge Columns Extract Parse From Text Statistics Standard Scientific Rounding Information Date Time Duration From Date & Time

Table: ReorderColumns(#"Changed Type4", {"Order ID", "Amount", "Profit", "Profit Margin", "Quantity", "Category"})

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name: Profit Status

If: Profit is less than 0 Then: Loss

Else If: Profit equals 0 Then: Break - Even

Else If: Profit is greater than 0 Then: Profit

OK Cancel

	Order ID	Amount	Profit	Profit Margin	Quantity	Category
19	B-25604	157.00	5.00	3.18%	9	Clothing

Transform Add Column View Tools Help

Transpose Reverse Rows Count Rows Data Type: Text Detect Data Type Rename Replace Values Fill Pivot Column Unpivot Columns Move Convert to List Split Column Merge Columns Extract Parse Statistics Standard Scientific Rounding Information Date Time Duration From Date & Time

Table.TransformColumnTypes(#"Reordered Columns1", {"Profit Status", type text})

	Amount	Profit	Profit Margin	Profit Status
1	1,275.00	-1,148.00	-90.04%	Loss
2	66.00	-12.00	-18.18%	Loss
3	8.00	-2.00	-25.00%	Loss
4	80.00	-56.00	-70.00%	Loss
5	168.00	-111.00	-66.07%	Loss
6	424.00	-272.00	-64.15%	Loss
7	2,617.00	1,151.00	43.98%	Profit
8	561.00	212.00	37.79%	Profit
9	119.00	-5.00	-4.20%	Loss
10	1,355.00	-60.00	-4.43%	Loss

○ Data Validation

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Manage Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Use First Row as Header Replace Values Transform

Queries [4]

List of Orders Order Details Sales target Order Data

Table.NestedJoin(#"List of Orders", {"Order ID"}, #"Order Details", {"Order ID"}, "Order Details", JoinKind.LeftOuter)

	Order ID	Order Date	CustomerName	State	City	Location
1	B-25601	01-Apr-18	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
2	B-25602	01-Apr-18	Pearl	Maharashtra	Pune	Pune, Maharashtra
3	B-25603	03-Apr-18	Jahan	Madhya Pradesh	Bhopal	Bhopal, Madhya Pradesh
4	B-25604	03-Apr-18	Divsha	Rajasthan	Jaipur	Jaipur, Rajasthan
5	B-25605	05-Apr-18	Kasheen	West Bengal	Kolkata	Kolkata, West Bengal
6	B-25606	06-Apr-18	Hazel	Karnataka	Bangalore	Bangalore, Karnataka



Visualization and Insights

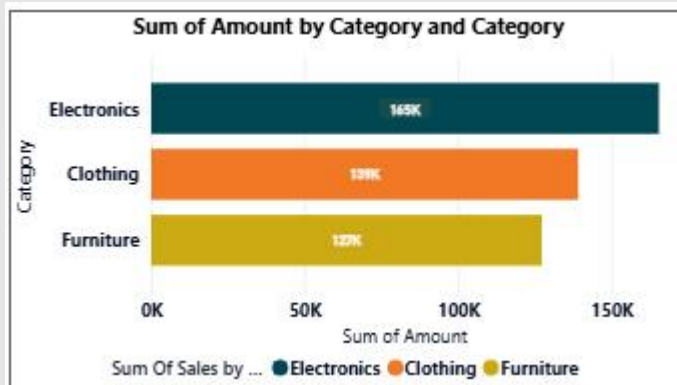
(Attach Screenshot of Visualizations)

Include screenshots of:

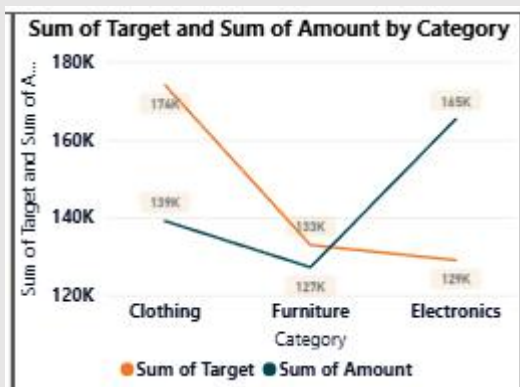
- **KPI Cards**



- **Bar Chart**



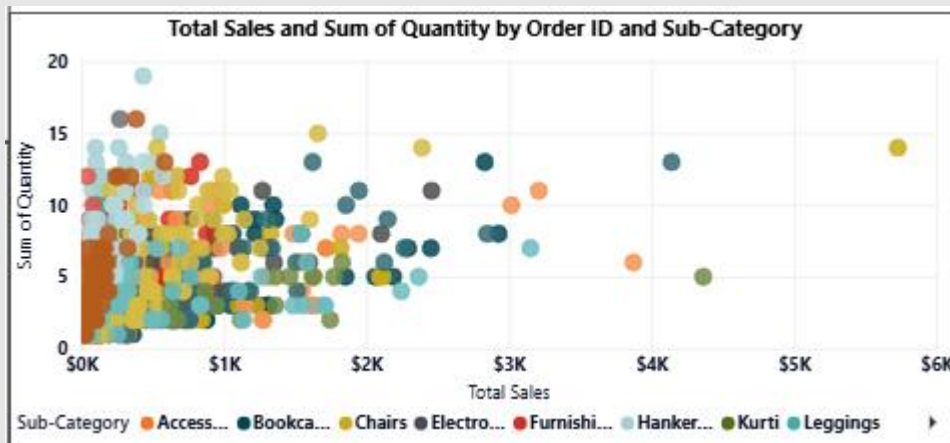
- **Line Chart**



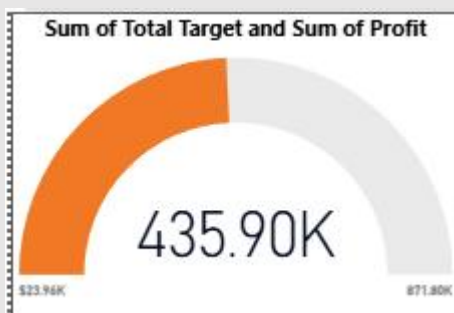
- **Pie/Donut Chart**



- Scatter Chart



- Gauge Chart



- Table

CustomerName	Total Sales
Yohann	\$1,314
Yogesh	\$5,421
Yash	\$791
Yaanvi	\$9,177
Wale	\$130
Vivek	\$114
Vishakha	\$4,836
Vipul	\$16
Vini	\$2,139
Vineet	\$666
Vikash	\$1,350
Total	\$431,502

Key Insights:

- Highest revenue-generating category identified.
- Top-performing products contributed major sales volume.
- Profit varied across categories and regions.
- Seasonal/monthly trends observed in customer purchasing behavior.

Report / Dashboard Screenshot

Insert final dashboard screenshot here.

Dashboard Components:

- KPI Summary
- Sales Performance
- Profit Analysis
- Product Performance
- Interactive Filters



Analysis

1. Descriptive Analysis

Purpose:

Summarize historical data and show overall performance.

- Total Sales achieved: **\$165.267K**
- Total Profit: **435.90K**
- Category-wise sales:
 - Electronics → Highest sales (~156K)
 - Clothing → ~136K
 - Furniture → ~121K
- Customer-wise sales table shows top contributing customers.

Key Insight:

Electronics generated the highest revenue compared to other categories.

2. Diagnostic Analysis

Purpose:

Identify reasons behind performance.

- Compare **Sales vs Target by Category**
 - Electronics exceeded target.
 - Furniture showed lower sales and missed target.
- Scatter chart (**Sales vs Quantity**) shows:
 - Higher quantity orders do not always generate higher sales.
 - Some products generate high sales with fewer quantities.

Key Insight:

Category performance differences and product demand influence overall revenue.

3. Predictive Analysis

Purpose:

Use trends to forecast future outcomes.

- Electronics sales may continue growing if demand remains stable.
- Furniture could continue underperforming unless strategy changes.
- High-performing customers may contribute more future revenue.

Prediction:

If current sales trends continue, Electronics is expected to remain the leading category.

4. Prescriptive Analysis

Purpose:

Recommend actions for improvement.

Recommendations:

- Increase inventory and promotions for Electronics.
- Introduce discounts or campaigns for Furniture.
- Focus customer retention programs on top buyers.
- Optimize product mix based on quantity vs sales relationship.

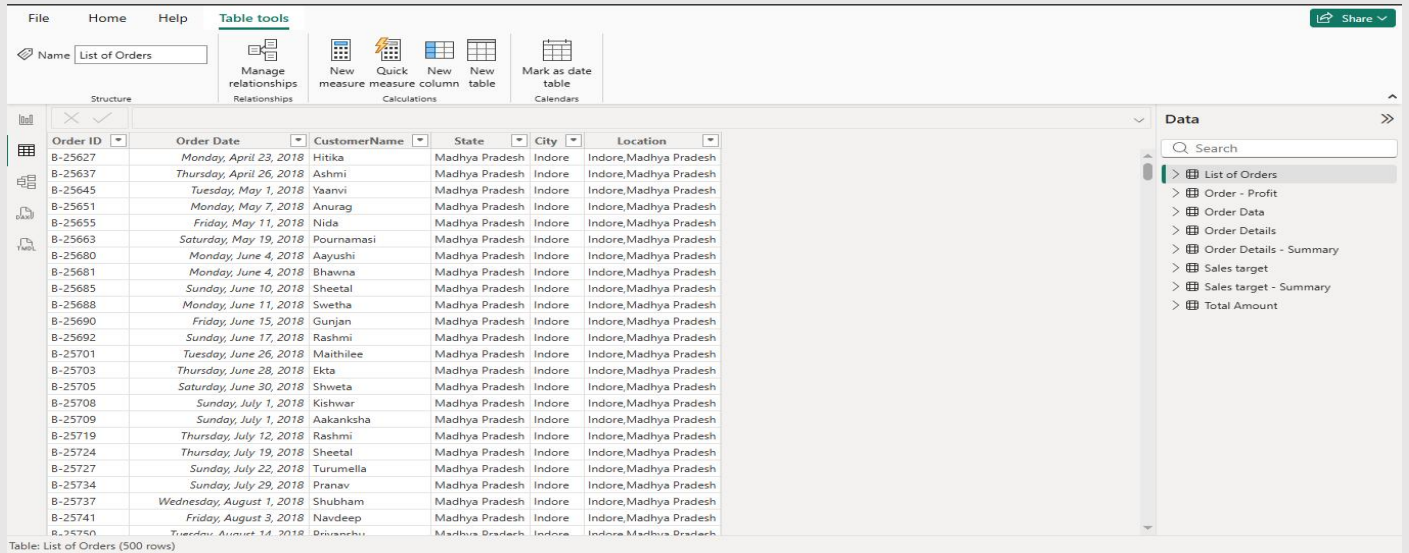
Action Plan:

- Maintain high-performing categories.
- Improve low-performing segments.
- Use customer insights for targeted marketing.

Conclusion

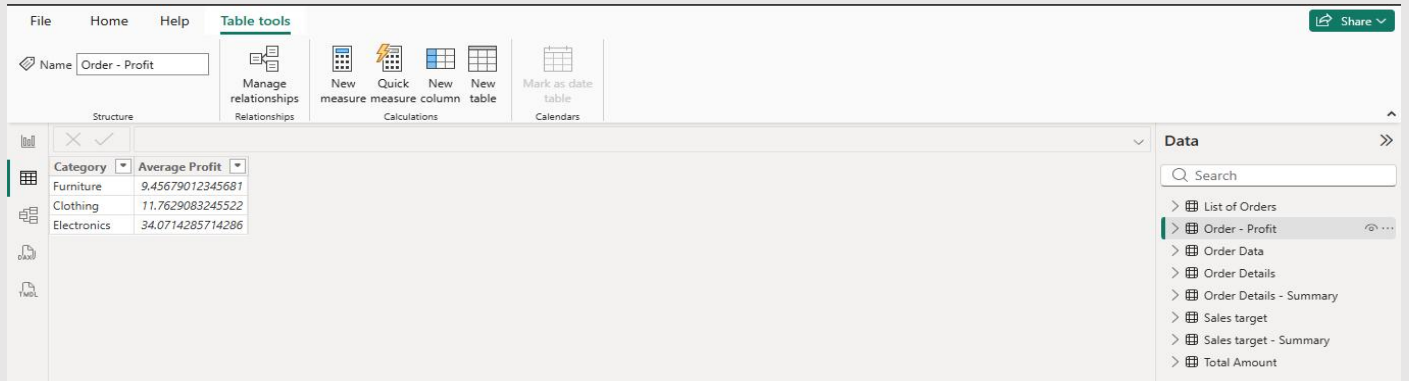
This project transformed raw data into meaningful business insights using Power BI. Through data cleaning, modeling, visualization, and analytical techniques, trends and performance drivers were identified. The dashboard supports data-driven decision-making by helping stakeholders monitor KPIs, understand customer behavior, and improve business performance

1. LIST OF ORDER



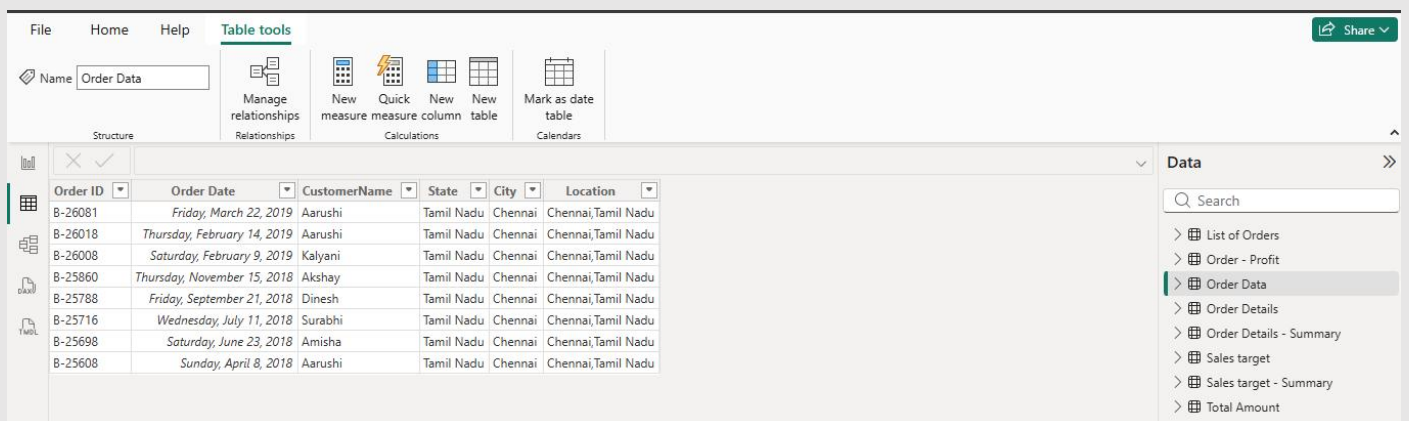
Order ID	Order Date	CustomerName	State	City	Location
B-25627	Monday, April 23, 2018	Hitika	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25637	Thursday, April 26, 2018	Ashmi	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25645	Tuesday, May 1, 2018	Yaanvi	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25651	Monday, May 7, 2018	Anurag	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25655	Friday, May 11, 2018	Nida	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25663	Saturday, May 19, 2018	Pournamasi	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25680	Monday, June 4, 2018	Aayushi	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25681	Monday, June 4, 2018	Bhawna	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25685	Sunday, June 10, 2018	Sheetal	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25688	Monday, June 11, 2018	Swetha	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25690	Friday, June 15, 2018	Gunjan	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25692	Sunday, June 17, 2018	Rashmi	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25701	Tuesday, June 26, 2018	Maithilee	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25703	Thursday, June 28, 2018	Ekta	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25705	Saturday, June 30, 2018	Shweta	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25708	Sunday, July 1, 2018	Kishwar	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25709	Sunday, July 1, 2018	Aakanksha	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25719	Thursday, July 12, 2018	Rashmi	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25724	Thursday, July 19, 2018	Sheetal	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25727	Sunday, July 22, 2018	Turumella	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25734	Sunday, July 29, 2018	Pranav	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25737	Wednesday, August 1, 2018	Shubham	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25741	Friday, August 3, 2018	Navdeep	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25750	Tuesday, August 14, 2018	Divyanshu	Madhya Pradesh	Indore	Indore,Madhya Pradesh

2. ORDER - PROFIT



Category	Average Profit
Furniture	9.45679012345681
Clothing	11.7629083245522
Electronics	34.0714285714286

3. ORDER DATA



Order ID	Order Date	CustomerName	State	City	Location
B-26081	Friday, March 22, 2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu
B-26018	Thursday, February 14, 2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu
B-26008	Saturday, February 9, 2019	Kalyani	Tamil Nadu	Chennai	Chennai,Tamil Nadu
B-25860	Thursday, November 15, 2018	Akshay	Tamil Nadu	Chennai	Chennai,Tamil Nadu
B-25788	Friday, September 21, 2018	Dinesh	Tamil Nadu	Chennai	Chennai,Tamil Nadu
B-25716	Wednesday, July 11, 2018	Surabhi	Tamil Nadu	Chennai	Chennai,Tamil Nadu
B-25698	Saturday, June 23, 2018	Amisha	Tamil Nadu	Chennai	Chennai,Tamil Nadu
B-25608	Sunday, April 8, 2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu

4. ORDER DETAILS

FileHomeHelpTable tools

NameOrder Details

Manage relationships

New measure

Quick measure

New column

New table

Mark as date table

Calendars

Structure

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Margin
B-25603	12	\$1	2	Clothing	Hankerchief	0.0833333333333333	Profit	0.0833333333333333
B-25608	257	\$23	5	Clothing	Hankerchief	0.0894941634241245	Profit	0.0894941634241245
B-25615	68	\$20	5	Clothing	Hankerchief	0.294117647058824	Profit	0.294117647058824
B-25616	42	\$12	5	Clothing	Hankerchief	0.285714285714286	Profit	0.285714285714286
B-25624	26	\$12	3	Clothing	Hankerchief	0.461538461538462	Profit	0.461538461538462
B-25625	97	\$29	2	Clothing	Hankerchief	0.298969072164948	Profit	0.298969072164948
B-25635	40	\$16	3	Clothing	Hankerchief	0.4	Profit	0.4
B-25638	154	\$39	3	Clothing	Hankerchief	0.253246753246753	Profit	0.253246753246753
B-25654	34	\$12	3	Clothing	Hankerchief	0.352941176470588	Profit	0.352941176470588
B-25656	6	\$3	1	Clothing	Hankerchief	0.5	Profit	0.5
B-25656	56	\$18	2	Clothing	Hankerchief	0.321428571428571	Profit	0.321428571428571
B-25670	24	\$1	2	Clothing	Hankerchief	0.0416666666666667	Profit	0.0416666666666667
B-25670	14	\$2	1	Clothing	Hankerchief	0.142857142857143	Profit	0.142857142857143
B-25730	18	\$8	2	Clothing	Hankerchief	0.444444444444444	Profit	0.444444444444444
B-25751	32	\$7	3	Clothing	Hankerchief	0.21875	Profit	0.21875
B-25757	106	\$15	7	Clothing	Hankerchief	0.141509433962264	Profit	0.141509433962264
B-25757	14	\$5	1	Clothing	Hankerchief	0.357142857142857	Profit	0.357142857142857
B-25757	17	\$7	3	Clothing	Hankerchief	0.411764705882353	Profit	0.411764705882353
B-25771	148	\$59	3	Clothing	Hankerchief	0.398648648648649	Profit	0.398648648648649
B-25783	26	\$2	2	Clothing	Hankerchief	0.0769230769230769	Profit	0.0769230769230769
B-25784	15	\$4	1	Clothing	Hankerchief	0.266666666666667	Profit	0.266666666666667
B-25793	18	\$1	3	Clothing	Hankerchief	0.0555555555555556	Profit	0.0555555555555556
B-25797	31	\$1	2	Clothing	Hankerchief	0.032258064516129	Profit	0.032258064516129
B-25800	45	\$12	7	Clothing	Hankerchief	0.266666666666667	Profit	0.266666666666667

Table: Order Details (1,500 rows)

Data

Search

List of Orders

Order - Profit

Order Data

Order Details

Order Details - Summary

Sales target

Sales target - Summary

Total Amount

5. ORDER DETAILS - SUMMARY

FileHomeHelpTable tools

NameOrder Details - Su...

Manage relationships

New measure

Quick measure

New column

New table

Mark as date table

Calendars

Structure

Order ID	Order Count
B-25605	1
B-25606	1
B-25607	1
B-25611	1
B-25612	1
B-25613	1
B-25615	1
B-25617	1
B-25619	1
B-25620	1
B-25622	1
B-25624	1
B-25627	1
B-25629	1
B-25631	1
B-25632	1
B-25634	1
B-25636	1
B-25637	1
B-25639	1
B-25641	1
B-25642	1
B-25644	1
B-25646	1

Table: Order Details - Summary (500 rows)

Data

Search

List of Orders

Order - Profit

Order Data

Order Details

Order Details - Summary

Sales target

Sales target - Summary

Total Amount

6. SALES TARGET

File Home Help Table tools Share

Name: Sales target

Structure: Manage relationships Relationships

Calculations: New measure, Quick measure column, New table, Mark as date table

Calendars

Month of Order Date	Category	Target	Target Achievement
Saturday, April 18, 2026	Furniture	10400	1222.89%
Monday, May 18, 2026	Furniture	10500	1211.25%
Thursday, June 18, 2026	Furniture	10600	1199.82%
Saturday, July 18, 2026	Furniture	10800	1177.60%
Tuesday, August 18, 2026	Furniture	10900	1166.80%
Friday, September 18, 2026	Furniture	11000	1156.19%
Sunday, October 18, 2026	Furniture	11100	1145.77%
Wednesday, November 18, 2026	Furniture	11300	1125.50%
Friday, December 18, 2026	Furniture	11400	1115.62%
Monday, January 19, 2026	Furniture	11500	1105.92%
Thursday, February 19, 2026	Furniture	11600	1096.39%
Thursday, March 19, 2026	Furniture	11800	1077.81%
Saturday, April 18, 2026	Clothing	12000	1158.78%
Monday, May 18, 2026	Clothing	12000	1158.78%
Thursday, June 18, 2026	Clothing	12000	1158.78%
Saturday, July 18, 2026	Clothing	14000	993.24%
Tuesday, August 18, 2026	Clothing	14000	993.24%
Friday, September 18, 2026	Clothing	14000	993.24%
Sunday, October 18, 2026	Clothing	16000	869.09%
Wednesday, November 18, 2026	Clothing	16000	869.09%
Friday, December 18, 2026	Clothing	16000	869.09%
Monday, January 19, 2026	Clothing	16000	869.09%
Thursday, February 19, 2026	Clothing	16000	869.09%
Thursday, March 19, 2026	Clothing	16000	869.09%

Table: Sales target (36 rows)

Data: Search

- List of Orders
- Order - Profit
- Order Data
- Order Details
- Order Details - Summary
- Sales target
- Sales target - Summary
- Total Amount

7. SALES TARGET - SUMMARY

File Home Help Table tools Share

Name: Sales target - Sum...

Structure: Manage relationships Relationships

Calculations: New measure, Quick measure column, New table, Mark as date table

Calendars

Month	Total Target
April	31400
May	31500
June	31600
July	33800
August	33900
September	34000
October	36100
November	36300
December	36400
January	43500
February	43600
March	43800

Data: Search

- List of Orders
- Order - Profit
- Order Data
- Order Details
- Order Details - Summary
- Sales target
- Sales target - Summary
- Total Amount

8. TOTAL AMOUNT

File Home Help Table tools Share

Name: Total Amount

Structure: Manage relationships Relationships

Calculations: New measure, Quick measure column, New table, Mark as date table

Calendars

Sub-Category	Total Amount
Bookcases	56861
Stole	18546
Hankerchief	14608
Electronic Games	39168
Phones	46119
Saree	53511
Trousers	30039
Chairs	34222
Kurti	3361
T-shirt	7382
Shirt	7555
Leggings	2106
Tables	22614
Printers	58252
Accessories	21728
Furnishings	13484
Skirt	1946

Data: Search

- List of Orders
- Order - Profit
- Order Data
- Order Details
- Order Details - Summary
- Sales target
- Sales target - Summary
- Total Amount